

Tutorial No.10

Q.1) Find the Algebraic Multiplicity, Geometric Multiplicity and Eigen space of each Eigen value of the following matrices:

$$1. A = \begin{bmatrix} -9 & 2 & 6 \\ 5 & 0 & -3 \\ -16 & 4 & 11 \end{bmatrix}$$

$$2. A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$$

$$3. A = \begin{bmatrix} 7 & -2 & 1 \\ -2 & 10 & -2 \\ 1 & -2 & 7 \end{bmatrix}$$

Q.2) Diagonalize the following matrices

$$1. A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$2. A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

Q.3) Convert quadratic form, $Q(x_1, x_2) = 5x_1^2 - 4x_1x_2 + 8x_2^2$ into canonical form using orthogonal diagonalization and hence identify the conic section for, $Q(x_1, x_2) = 1$